Wealth Advisory Agent — Full Product & Data Documentation

Theme: Financial Participant
Track: Topic B — Wealth Advisory Agent
Core Product: An AI agent that profiles a client, studies live markets, recommends specific ETFs/bonds/stocks, explains why, and executes the trade inside a free paper-trading sandbox.

Scoring Alignment

Dimension	Weight	How This Product Wins
Task Completion	40%	Complete end-to-end loop: risk profiling → product comparison → recommendation → execution → portfolio tracking. Not a chatbot. A working agent.
Security & Compliance	30%	Runtime compliance guard checks every recommendation and trade against suitability and concentration rules. Full audit log.
Innovation & Interaction	30%	Behavioral bias detection + live market data + explainable recommendations + real trade execution.
Challenge Bonus	+0–5	Optional: log trade hashes on-chain for immutable audit trail.

Open Data Sources by Feature

No manual data entry. Everything below is API-driven and free.

Data Need	Source	What It Provides	Key / Cost
Stocks, ETFs, indices, options, FX, commodities	fin-data-mcp (GitHub, 83 no-key tools)	One-stop MCP server with live API requests across 13 market-data categories. Equities have two independent no-key backends (Yahoo + Cboe).	No API key needed for 8...
US Treasury yields & bond data	mcp-treasury (GitHub)	Daily Treasury yield curve rates (bill, note, bond), auction results, national debt, exchange rates.	Free API, no key required
ETF holdings & fund allocation	FTMarkets / ftgo (PyPI)	ETF/fund holdings, asset allocation, sector breakdowns, historical OHLCV for stocks and ETFs.	Free Python library
Macro time series (34 series)	CalcFi Open Data (Zenodo/HuggingFace)	Treasury yields, interest rates, CPI, unemployment, energy prices, FX, crypto, FDIC deposit rates. CC-BY-4.0.	Free, daily-refreshed mi...
Stock/ETF screener data	yfscreen (R/Python)	Yahoo Finance screener: filter ETFs by expense ratio, historical performance, and other attributes.	Free
Broad financial datasets for ML	finmind (PyPI)	50+ datasets: US stock daily/minute, oil, gold, G7 exchange rates, interest rates.	Free, open source

Data Need	Source	What It Provides	Key / Cost
Behavioral finance bias data	Kuvera-PersonalFinance-V2.1 (HuggingFace)	Behaviorally-grounded reasoning chains covering loss aversion, overconfidence, and other biases.	Free dataset
Psychological drivers of economic decisions	HuggingFace dataset	Global evidence on psychological factors driving economic decision-making (1990–2025).	Free dataset
Financial sentiment	Stocktwits sentiment dataset (Zenodo)	953,209 messages classified by FinTwitBERT-sentiment.	Free dataset
KYC/AML synthetic data	VynFi AML (HuggingFace)	748K synthetic banking transactions with AML/SAR labels and KYC/AML typologies.	Apache-2.0
Trade execution (paper)	Alpaca Paper Trading API	Real-time simulation against live NBBO quotes. Place buy/sell orders, track portfolio.	Free paper account
Portfolio optimization	skfolio (PyPI)	Portfolio optimization built on scikit-learn. Efficient frontier, risk models.	Open-source BSD

Feature 1: Dynamic Conversational Risk Profiling

Selling Point: "A living risk profile that understands you, not just your age."
Most solutions use a static 10-question form. Yours adapts follow-up questions based on user responses, probes emotional drivers, and builds a holistic risk profile. This is the trust foundation for everything else.

Feasibility: High.

Implementation Entails:

- Conversational flow using an LLM with a state machine.
- Extract risk parameters from dialogue and store in a typed state object.
- Use Kuvera-PersonalFinance-V2.1 or the psychological drivers dataset to ground the bias detection later.
- KYC/AML questions can be seeded from VynFi AML typologies.

Open Data Used:

- Behavioral finance bias dataset (HuggingFace)
- Psychological drivers dataset (HuggingFace)
- KYC/AML synthetic data (VynFi)

Feature 2: Multi-Dimensional Product Comparison & Suitability Filtering

Selling Point: "Fiduciary-grade product matching with a rationale."
The agent doesn't just pick one product. It pulls live data on dozens of ETFs and bonds, scores them against the client's profile, and presents a transparent comparison: "We chose BND over AGG because it has lower fees and fits your 2-year horizon better."

Feasibility: High.

Implementation Entails:

- Pull live ETF data (expense ratio, holdings, performance) via FTMarkets or yfscreen.
- Pull bond data (yields, duration, credit rating) via mcp-treasury and CalcFi Open Data.
- Pull broad market data via fin-data-mcp.
- Comparison engine scores products against risk profile, horizon, and liquidity needs.
- Compliance Agent filters out unsuitable products before they are shown.

Open Data Used:

- FTMarkets / ftgo for ETF holdings and allocation
- mcp-treasury for bond yields and auction results
- fin-data-mcp for broad market data
- CalcFi Open Data for macro context

Feature 3: Explainable Asset Allocation & Portfolio Construction

Selling Point: "A portfolio you can understand and trust."
The system generates a recommended allocation and provides layered explanations: a simple summary for the user and a technical rationale for a human advisor. This directly addresses the "black box" problem.

Feasibility: High.

Implementation Entails:

- Use skfolio or cvxportfolio to run Modern Portfolio Theory (MPT) and generate an efficient frontier.
- Use SHAP/LIME to explain why the model allocated specific weights.
- The Explainability Agent converts technical outputs into plain-English narrative.

Open Data Used:

- skfolio for optimization
- fin-data-mcp for asset price history
- CalcFi Open Data for risk-free rate and macro context

Feature 4: Behavioral Finance & Bias Detection

Selling Point: "A psychologically-aware financial co-pilot."
Most robo-advisors ignore behavioral finance. Yours notices when a user is about to panic-sell or chase hype, and gently intervenes: "I see you're checking your portfolio daily and reacting to short-term dips. That's loss aversion. Let's revisit your long-term goals."

Feasibility: Medium-High.

Implementation Entails:

- Use the Kuvera-PersonalFinance-V2.1 dataset to fine-tune or prompt an LLM for bias detection.
- Analyze dialogue for phrases indicating overconfidence, loss aversion, or herding.
- Generate personalized debiasing interventions.
- Track behavioral patterns over time in a "behavioral memory."

Open Data Used:

- Kuvera-PersonalFinance-V2.1 (HuggingFace)
- Psychological drivers dataset
- Stocktwits sentiment dataset for market mood context

Feature 5: Runtime Compliance & Safeguards

Selling Point: "Compliance-by-design, with a verifiable audit trail."
You are not just saying your system is compliant. You are demonstrating a runtime mechanism that enforces it. Every recommendation and trade passes through a compliance guard before it reaches the user.

Feasibility: High.

Implementation Entails:

- Rule engine or LLM-based guard with explicit compliance contracts (e.g., "Recommendations must be suitable for the client's risk profile," "No promises of guaranteed returns").
- Orchestrator routes every output through this agent before presentation.
- Every check, result, and final output is logged to an immutable audit trail.
- Permission model: non-discretionary — the agent can only advise and execute on explicit user approval.

Open Data Used:

- KYC/AML synthetic data (VynFi) for rule seeding
- Regulatory rule references from MAS SAFR framework (public documentation)

Feature 6: Real-Time Market Monitoring & Advisory Dialogue

Selling Point: "Your portfolio's health, explained monthly."
The agent simulates a "monthly review" where it checks portfolio performance against the plan, alerts the user to drift, and explains the reasons in plain English.

Feasibility: Medium.

Implementation Entails:

- Scheduled script or "simulate time passing" button.
- Pull live prices via fin-data-mcp and compare current allocation to target.
- Generate natural-language "Portfolio Health Report."
- Use CalcFi Open Data for macro context in explanations.

Open Data Used:

- fin-data-mcp for live prices
- CalcFi Open Data for macro context

Feature 7: Real Trade Execution (The "Action" Layer)

Selling Point: "An AI advisor that doesn't just talk — it trades."
This is your biggest differentiator. The AI says "buy BND" and then actually places the order in a free paper-trading account. Judges will see the trade fill, the shares appear, and the portfolio update live.

Feasibility: High.

Implementation Entails:

- Integrate Alpaca Paper Trading API.
- Use the Python SDK (alpaca-py) to place market orders.
- Paper trading base URL: https://paper-api.alpaca.markets.
- After user clicks "Invest," the Execution Module places the order and returns the fill confirmation.
- Log the transaction hash for the audit trail (and optionally on-chain for bonus points).

Open Data Used:

- Alpaca Paper Trading API (free account)
- Alpaca Market Data API for real-time quotes

Feature 8: Dashboard & Portfolio Tracking

Selling Point: "See your money grow, and understand every move."
A clean dashboard showing current holdings, performance, and next steps.

Feasibility: High.

Implementation Entails:

- Pull portfolio positions from Alpaca API.
- Visualize allocation vs. target allocation.
- Show performance chart using historical data from fin-data-mcp.
- Display compliance checks and audit log entries.

Open Data Used:

- Alpaca API for positions and performance
- fin-data-mcp for historical charts

The One-Sentence Pitch

"We built an AI wealth advisor that studies live markets, recommends specific ETFs and bonds with plain-English explanations, catches your emotional mistakes, proves it follows the rules, and lets you execute the trade in one click — inside a free, safe sandbox."

Final Checklist for the Hackathon

Submission Item	What to Include
Technical documentation	Multi-agent architecture diagram; data source table; compliance rule list.

Submission Item	What to Include
Presentation deck	Lead with the "talk vs. action" differentiator. Show live data sources and execution.
Demo video (≤5 min)	Screen capture: profiling → recommendation → explanation → trade execution → portfolio update.
Source code	GitHub repo with README, deployment instructions, test cases.
Security self-assessment report	Non-discretionary permission model; known-risk list (LLM hallucination, data privacy); mitigations.
Execution evidence	Alpaca paper-trade operation logs; screenshots of filled orders.